Appendices A-D and F-K (for draft 6)

Draft 6, 2026-10-16. Main text: `docs/paper/DRAFT6_workshop.md`. *Appendix E (theory) is the separate file* `docs/paper/APPENDIX_theory_v3.md`; *its statements are cited here by their labels there (A0, B0, T1 = Lemma 1, P1-P7, Remark R). Every number below is copied from the file named beside it; a CORRECTED or AUDIT block at the top of a results file supersedes its body, and was read first. Status words: registered (a pre-registered decision or declared secondary), post hoc (computed after the registered analysis; never changes a verdict), descriptive (a registered arm with no decision attached), exploratory (not registered). Nothing here is new computation: the tables are copies of committed analysis outputs, reformatted.*

Contents. A: models and recipes. B: floors and gates. C: per-seed tables and post hoc re-analyses. D: constructions. (E: theory, separate file.) F: Dyck-2 at matched depth. G: navigation in words. H: faithful CARoPE and CoPE. I: the toy. J: the clock side. K: depth.

A. Models and recipes

A1. Architecture (every navigation batch)

Source: `model.py` (`MapFormerWM`, `WMTransformerLayer`, `ActionToLieAlgebra`, `PathIntegrator`), `model_sign.py`, `model_wrap_exist.py`, `model_mono_baselines.py`, `model_baseline_rope.py`, and the trainers named in A5.

component	setting
layers	1 (appendix K also 4 real layers and one block looped x4; appendix F 1-4 layers; ROBUST torus 2 layers, d_model 256, C13)
heads, width	2 heads, d_model 128, d_head 64; FFN 4 x d_model with GELU; LayerNorm before attention and before the FFN; dropout 0.1 on attention weights, attention output and FFN output
rotary channels	32 two-dimensional blocks per head, 64 channels; every query and key coordinate is rotated
score	channel c contributes q_c^T R(theta_{t',c} - theta_{t,c}) k_c (MapWM: content Q, K rotated by the path angle; not additive)
step map	Delta = W_out W_in x, W_in: 128 -> r, W_out: r -> 64, no bias, r = 4 (_r4), one latent shared across the heads (MapFormer's is per head)
frequencies	omega_c = 2 pi (1 / Delta_max)^(i / 31) for block i = 0..31, the same ladder in each head; Delta_max = the grid size (torus 64, ring 32, line 256); omega is a free parameter (its sign is not constrained)
vocabulary	torus: 4 moves + 17 observation tokens (16 types + blank); ring and line: 2 moves + 17
parameters	~204k for the one-layer torus MapWM (`FACTORIAL_RESULTS.md`)

A2. Step parameterisations and arm names

name in the files	main-text name	step Delta (per channel, after W_out)	source
`Signed_r4` (= `Vanilla_r4`)	signed	W_out W_in x	`model.py`

name in the files	main-text name	step Delta (per channel, after W_out)	source
`Abs_r4`	|.|	|W_out W_in x|	`model_sign.py` mode "abs"
`Pos_r4`	softplus	softplus(W_out W_in x); softplus(0) = 0.693, so every token starts at about the same step	`model_sign.py` mode "pos"
`SignedAO_r4`, `AbsAO_r4` (ring); `SignedAO4_r4`, `AbsAO4_r4` (torus)	observation-step-free (AO) signed / |.|	as above for the move tokens; every other token's step is 0 by construction	`model_wrap_exist.py`
`RoPE`	index RoPE	canonical ladder base^(-2c / d_head), base 10000	`model_baseline_rope.py`
`CARoPEF`	CARoPE	appendix H	`model_mono_baselines.py`
`CoPEF` (and `_Hi`, `_Lo`, `_LoS` table starts)	CoPE	appendix H	`model_mono_baselines.py`

The |.| constraint is applied after W_out, on each of the 64 channel steps, so every channel is non-negative; applying it on the r-dimensional latent would leave W_out free to mix signs (`model_sign.py` docstring). It adds no parameters.

Initial move angles (Table 1 arms; `train_wrap_mech.py`). default: the seed's initialisation. *Same-direction start* (`_absmag`): |default angles| in the balanced gauge (appendix D2) -- the start |.| imposes. *Large, sums below pi* (`AO_mid`): a(+1), a(-1) iid U(0.9, 1.5) rad per channel. *Large, sums above pi* (`AO_high`, `SO_high`): iid U(1.7, 4.5) rad. Every other weight is the seed's default initialisation, identical across arms at a seed. Torus large starts (`train_torus_start.py`): the same distributions, iid per channel and move, balanced gauge (`TORUS_START_PREREG.md`). *Frozen move latents* (`_fz`, ACCOUNTS T2): W_in fixed and the move embeddings detached in the step path (they still train through the content path), so only W_out and omega move the angles.

A3. Optimisation (all from `train.train`, the trainers' argparse defaults and the drivers)

	torus (Secs. 4.1, 4.2, 4.4, 4.6; K)	ring and line (Secs. 4.1-4.6)
optimiser	AdamW, weight decay 0.05 (`train.train` default; ACCOUNTS T1 also 0), default betas	same
peak learning rate	1e-3	1e-3
schedule	per step: 5% linear warm-up, then cosine to 10% of the peak	same
batch	16 walks of T = 1024 moves (2,048 tokens, interleaved move / observation)	128 walks of T = 128 moves
batches per epoch, epochs	98, 900	98, 300 (RING_BUDGET also 900)
loss	cross-entropy on revisited observation positions only	same
data	3 parallel data workers (`--data-workers 3`)	same

	torus (Secs. 4.1, 4.2, 4.4, 4.6; K)	**ring and line (Secs. 4.1-4.6)**
attention kernel	SIGN_MATCHED: standard; GENERALITY, TORUS_START, FACTORIAL, ROBUST: `--fast-attn` (SDPA, TF32 off)	standard
recipe variants	ROBUST torus: 2 layers, d_model 256 (C13)	ROBUST ring: lr 3e-3 and 3.3e-4 (wd 0.05), and wd 0 (lr 1e-3) (C13)
training map	one per run (GridWorld seed = run seed); walks are runs of 1-10 repeated moves	redrawn for every walk (`environment_cancel`)

Cross-batch torus comparisons. Sec. 4.1 sets SIGN_MATCHED (full `Signed_r4` / `Abs_r4`, standard attention) beside GENERALITY and TORUS_START (observation-step-free `SignedAO4_r4` / `AbsAO4_r4`, `--fast-attn`). These differ in attention kernel and model class: `--fast-attn` agrees with the standard path to ~1e-6 in logits but is not bit-identical in training (dropout RNG). Only within-batch contrasts carry verdicts. ROBUST's torus cell shares GENERALITY's kernel and classes and differs in depth, width and seeds, so "a quarter of the one-layer gap" (Sec. 4.1) is out of batch but not across kernels.

Continuations (appendix D5): ring 300 epochs at lr 1e-3, schedule re-warmed, fresh optimiser state, walk stream offset 10^6 (`train_wrap_exist.py`); torus 300 epochs at lr 1e-4 (primary) or 1e-3 (descriptive), `--data-seed-offset 1000000` (`run_generality.sh`).

A4. Evaluation

- **Torus** (`eval_noise_refine.evaluate`): one held-out map (GridWorld seed 10000), 100 walks per run with numpy seed 1234 + run seed, revisit accuracy at T = 1024 (registered) and 2048 (out of distribution, no verdict), eval mode (no dropout). Revisit strata from `eval_rank_strata` (lag < 128, lag >= 128, wrap-only).
- **Ring and line** (`train_cancel.evaluate`): maps redrawn per walk (env seed 10000), 400 walks, numpy seed 10^6 for every run, accuracy at T = 128 (registered) and 4T (out of distribution).
- **Dropout re-score**: attention scaled by 1/(1 - p) at eval. A declared secondary in RING_DIAL, RING_BUDGET, WRAP_EXISTENCE, GENERALITY, MECHANISM and WRAP_COPE (no label changed; each results file states it), and applied to SIGN_MATCHED and TEXTWORLD by the 2026-10-04 audit (`docs/audits/2026-10-04/DROPOUT_RESCORE.md`; no verdict changed). ACCOUNTS, TORUS_START, GAUGE and ROBUST were not re-scored (ROBUST: the correction is invalid at 2 layers); FACTORIAL declared it for the 1-layer cells and does not report it.
- **Run classes** (`stats_core.classify_run`, per-epoch training loss): *SOLVED* if the mean of the final 5% of epochs is < 0.05; otherwise *STALLED* if the final-10% mean is within 5% of the preceding 10% mean, else *DESCENDING*.
- **Cancelling share** (Fig. 2, "wrap share"; draft 5's Fig. 1): among channels with |wrap(a(+1))| > 0.05 rad, the share with |wrap(a(+1) + a(-1))| < 0.1 rad (wrap-invariant). *Usage-weighted* (ACCOUNTS T1): channels weighted by query x key norm. *Opposite-sign share*: channels whose raw move angles have opposite signs. *Collapsed*: |a| <= 0.05 rad. The unweighted share excludes collapsed channels from its denominator, and unused channels dominate it (ACCOUNTS T1: 0.87 -> 0.18 unweighted vs 0.99 -> 0.98 weighted on the same runs); Fig. 2 is redrawn usage-weighted for draft 6.

A5. Batches

batch (results file)	main text	task	arms	seeds	trainer / driver	independent check before results
SIGN_MATCHED	4.1	torus	Signed_r4, Abs_r4, Pos_r4, RoPE	0-7	`train_variant` / `run_sign_matched.sh`	no: authors' pre-launch gate (`docs/audits/2026-09-27/gate_sign_matched.py`) audit after the run (2026-09-30), which reproduced its probe (no file records it as independent)
RING_DIAL	4.1	ring p_plus 0.5 / 0.75 / 0.9	Signed_r4, Abs_r4	30-37	`train_ring_dial` / `run_ring_dial.sh`	yes (Amendment 1)
RING_BUDGET	4.1	ring 0.5	Abs_r4 at 300 and 900 epochs	40-47	`train_ring_dial` / `run_ring_budget.sh`	yes
WRAP_EXISTENCE	4.1, 4.2	ring 0.5	SO, AO, constructed arms	600-607	`train_wrap_exist` / `run_wrap_existence.sh`	yes
GENERALITY	4.1, 4.2, 4.6	torus; ring 0.5	SO4, AO4, constructed; Signed_r4, CARoPEF, CoPEF, RoPE	700-707	`train_variant`, `train_ring_variant` / `run_generality.sh`	yes
MECHANISM	4.1, 4.3	ring 0.5	Table 1 arms	800-807	`train_wrap_mech` / `run_wrap_mech.sh`	yes
WRAP_COPE	4.6, H	ring 0.5	CoPEF table starts	900-907	`train_ring_variant` / `run_wrap_cope.sh`	yes
ACCOUNTS	4.2, 4.3, 4.6	ring 0.5; line	T1 continuations (600-607), T2 frozen latents (1000-1007), T4 line (1100-1115)	see arms	`train_wrap_exist`, `train_wrap_mech`, `train_ring_variant` / `run_accounts.sh`	yes
TORUS_START	4.4	torus	SO4, AO4, AO4_high, SO4_high, AO4_mid	1200-1207	`train_torus_start` / `run_torus_start.sh`	yes
GAUGE	4.5	ring 0.5, moves +1/-1 or +1/+2	SO, AO, RoPE	1300-1307	`train_gauge` / `run_gauge.sh`	yes

batch (results file)	main text	task	arms	seeds	trainer / driver	independent check before results
ROBUST	4.1, C13	ring 0.5 (+1/-1; four recipes); torus 2 layers d 256	SignedAO / AbsAO (ring), SignedAO4 / AbsAO4 (torus)	1400-1407	`train_gauge`, `train_variant` / `run_robust.sh`	yes (Amendment 1, `docs/audits/2026-10-16/robust_audit.md`)
CANCEL	B (index gate only)	ring p_plus 0.5 / 0.75 / 0.9 / 1.0	path 1 layer; index 1-3 layers	0-7	`train_cancel` / --	no
FACTORIAL	K	torus	path 1 layer; index 1, 4 layers, looped x4	500-503 (n = 4)	`train_variant` / `run_factorial.sh`	yes
TEXTWORLD	G, K	torus in words	path 1 layer; index 1, 2 layers	0-7	`train_textworld` / `run_textworld.sh`	yes (Amendment 1), bu seeds 0-1 are byte-identical to the pilot (CORRECTED block)
DYCK_MDEPTH	F, K	Dyck-2	see F	0-7	`train_dyck` / `run_dyck_mdepth.sh`	no: predates the independent checks
CLOCK_SIDE, CLOCK_SPEED, CLOCK_SPEED_REP	J	k-back recency	Signed_r4, Abs_r4, CARoPE_r4 (CLOCK_SIDE also Pos_r4, RoPE)	200-207; 210-225; 230-245	`run_clock_side.sh`, `run_clock_speed*.sh`	yes

The independent check is an automated code-verification agent: a separately launched, read-only AI code-review session given the pre-registration and the code but no results (none exist at that point), which may run smoke tests on seeds outside the batch and whose findings are written into the pre-registration as an amendment before launch (example: `docs/audits/2026-10-16/robust_audit.md`). It is independent of the session that wrote the code, not of the project. The main text uses CANCEL only through GENERALITY's checked index value (0.730); CANCEL's own numbers appear in B, labelled.

B. Floors and gates

A floor is the accuracy of the best map-free predictor on the same kind of evaluation walk. Every value is from the file named; rows marked post hoc were computed for this draft (2026-10-16); the rest are copied.

task	predictor	value	computed on	source
torus T = 1024	always predict blank		same generator and	`docs/audits/2026-10-05/score_rank_floors_out.txt`

task	predictor	value	computed on	source
		0.507 (per stream seed 0.497-0.511)	protocol as A4 (held-out map 10000, numpy seed 1234 + s, 100 walks) for s = 10-25	
torus T = 1024	best token n-gram, orders 0-5, fit on the training maps	0.576 (0.568-0.582)	same	same
torus T = 1024	retrace (copy the observation 2j steps back while a run reverses the previous run)	0.843 (0.833-0.848)	same	same; `SCORE_RANK_PREREG.md`
torus T = 1024	blank / n-gram / retrace on the SIGN_MATCHED and GENERALITY walks (post hoc)	0.506 / 0.575 / 0.843 and 0.508 / 0.578 / 0.843	each batch's own eval stream	`docs/audits/2026-10-16/sign_floors_out.txt`, `generality_floors_out.txt`
torus T = 1024, 2 layers d 256 (ROBUST)	blank / n-gram / retrace (post hoc)	0.509 / 0.578 / 0.844; every run of both arms is above retrace (lowest |.| 0.910)	ROBUST's own eval stream, seeds 1400-1407	`docs/audits/2026-10-16/robust_floors_out.txt`
torus T = 128 (MapFormer's length; PAPER2X2 eval walks)	blank / n-gram / retrace (post hoc)	0.507 / 0.598 / 0.934 (retrace per seed 0.9277-0.9455; no wrap-only revisits at this length); index RoPE (0.805) is below retrace, path r = 2 (0.971) just above	PAPER2X2's eval stream, seeds 0-7	`docs/audits/2026-10-16/paper2x2_T128_floors_out.txt` (`sign_floors.py` with `FLOOR_T=128`; n-gram still fit on T = 1024 walks of maps 10-25)
ring p_plus 0.5 / 0.75 / 0.9	retrace (post hoc)	0.860 / 0.758 / 0.612	2000 walks, numpy seed 4242	`docs/audits/2026-10-16/ring_floors_out.txt`
torus T = 1024	best constant, on each batch's own evaluation walks (identical across arms at a seed)	SIGN_MATCHED 0.501-0.512 per seed (mean 0.506); GENERALITY 0.504-0.518 (0.508);	the batch's registered eval stream	`SIGN_MATCHED_STRATA.json`, `GENERALITY_TORUS_SO4_STRATA.json`, `FACTORIAL_L1_STRATA.json` ('all' stratum, field `floor`)

task	predictor	value	computed on	source
		FACTORIAL 0.505-0.515 (0.509)		
ring p_plus 0.5 / 0.75 / 0.9	better of best constant and n-gram (orders 1-5) over preceding tokens	0.554 / 0.544 / 0.520	1000 held-out walks (env seed 10000, numpy seed 1); n-gram fit on 3000 walks of map seed 0	`CANCEL_PREREG.md`; `docs/audits/2026-09-27/gate_cancel.py`
ring p_plus 0.5 / 0.75 / 0.9	fixed-lag copy (exactly one lap back)	0.284 / 0.375 / 0.626	same	same
ring p_plus 0.5	constant predictor, per revisit stratum	0.49-0.50	the registered ring eval stream (400 walks, numpy seed 10^6)	`docs/paper/figs/FIGS_NOTES.md`
line (256 cells)	predict blank	0.503	ACCOUNTS T4 eval stream	`ACCOUNTS_RESULTS.md`
ring, moves +1/+2	predict blank	~0.50	GAUGE eval stream	`GAUGE_RESULTS.md`
torus in words	best constant / reversal-copy rule	0.505 / 0.597	the batch's eval set	`TEXTWORLD_RESULTS.md`
Dyck-2 L32 D12	best A2f floor (n-grams of order 1-6 fit on the D12 training distribution; stack-free)	0.594	gate G4	`DYCK_MDEPTH_PREREG.md`, `DYCK_MDEPTH_RESULTS.md`

Own-stream torus floors (post hoc, 2026-10-16). The n-gram and retrace floors were first computed on stream seeds 10-25 only. Recomputed on each batch's own evaluation walks with the same script (`docs/audits/2026-10-16/sign_floors.py`, a copy of `score_rank_floors.py` with the stream seeds as a parameter; n-gram still fit on the 16 training maps 10-25): SIGN_MATCHED (seeds 0-7) blank 0.506, n-gram 0.575, retrace 0.843 (per seed 0.837-0.849); GENERALITY (700-707) blank 0.508, n-gram 0.578, retrace 0.843 (`sign_floors_out.txt`, `generality_floors_out.txt` there). Against its own seed's retrace floor, SIGN_MATCHED |.| falls below on 3/8 runs (0.637, 0.690, 0.717), softplus on 4/8, index RoPE on 8/8, signed on 0/8; GENERALITY's observation-step-free |.| (AO4, mean 0.880) is above the mean retrace floor. So the unsolved barrier models' torus accuracy is, on average, at or below what a map-free local rule reaches; the gap to the signed model (0.998) and SOLVED are the evidence that the map was not found.

Ring retrace floor (post hoc, 2026-10-16). The same retrace rule on the ring (`docs/audits/2026-10-16/ring_floors.py`, 2000 walks per p_plus, scored on revisits as `train_cancel.evaluate`): 0.860 / 0.758 / 0.612 at p_plus 0.5 / 0.75 / 0.9 (blank 0.497-0.499; share of revisits inside a retrace 0.72 / 0.52 / 0.23). It is a stronger map-free floor than the registered 0.554 / 0.544 / 0.520 at p_plus 0.5 and 0.75. The ring |.| models (0.95-0.97 at p_plus 0.5) are above it. For moves +1/+2 (GAUGE) no move reverses another and retrace reduces to predicting blank (~0.50).

Torus at T = 128 (post hoc, 2026-10-16). The draft cites no T = 128 torus number; the row is for readers who know MapFormer's T = 128 results. At that length almost every revisit is local, and the map-free retrace rule scores 0.934, so a T = 128 torus accuracy below ~0.93 is not evidence of a map (PAPER2X2's converged index RoPE, 0.805, is below it; the path model's 0.971 is above it by 0.04).

Gates. Each task was gated before any GPU run (rule 11): the floors above call the task code. Whether a task needs a map at all is read from the in-batch one-layer index RoPE model, which cannot integrate content-dependent steps: torus 0.722-0.731 (GENERALITY, SIGN_MATCHED), ring p_plus 0.5 / 0.75 / 0.9 0.710 / 0.677 / 0.824 on CANCEL's six fresh seeds (0.717 / 0.676 / 0.825 with its two pilot seeds; CANCEL had no independent pre-launch check; `CANCEL_RESULTS.md` CORRECTED block) and 0.730 (GENERALITY, the value the main text uses), line 0.772 (ACCOUNTS T4), moves +1/+2 0.649 (GAUGE). Each is above its blank and n-gram floors and below the signed path model (1.000 or 0.998); on the torus and on the ring at p_plus 0.5 / 0.75 it is below the retrace floor.

C. Per-seed tables and post hoc re-analyses

Accuracy is at the training length (torus T = 1024 on the held-out map; ring T = 128), eval mode. S / SOLV = SOLVED, STAL = STALLED, DESC = DESCENDING; a number in brackets is the final-5% training loss.

C1. Torus sign batch (Sec. 4.1; `SIGN_MATCHED.json`, `SIGN_MATCHED_ANALYSIS.txt`)

seed	signed	|.|	softplus	index RoPE
0	1.000 SOLV (0.003)	0.931 DESC (0.201)	0.590 STAL (1.250)	0.730 STAL (0.983)
1	0.997 SOLV (0.003)	0.917 DESC (0.314)	0.527 STAL (1.510)	0.729 STAL (0.939)
2	1.000 SOLV (0.005)	0.868 STAL (0.492)	0.630 STAL (1.084)	0.720 STAL (0.931)
3	1.000 SOLV (0.008)	0.637 STAL (1.043)	0.972 DESC (0.158)	0.739 STAL (0.910)
4	1.000 SOLV (0.004)	0.690 DESC (0.962)	0.674 STAL (1.009)	0.737 STAL (0.928)
5	0.990 SOLV (0.003)	0.888 STAL (0.446)	0.935 DESC (0.295)	0.744 STAL (0.885)
6	1.000 SOLV (0.004)	0.922 STAL (0.349)	0.922 DESC (0.345)	0.722 STAL (0.956)
7	1.000 SOLV (0.004)	0.717 STAL (0.834)	0.995 SOLV (0.049)	0.728 STAL (0.976)
mean	0.998	0.821	0.781	0.731

Table C1. Registered: |.| - signed -0.177 (MDE ~0.14, permutation p 0.0002; SOLVED 0/8 vs 8/8, Fisher p 0.0002); softplus - signed -0.218 (p 0.0003; 1/8 vs 8/8, p 0.0014). r(final

loss, accuracy) -0.992 over 32 runs. Seed 5's signed run (0.9896) is the one below 0.99 under the accuracy criterion of C10.

C2. Ring dial (Sec. 4.1; `RING_DIAL_VERDICTS.json`)

seed	0.5 signed	0.5 |.|	0.75 signed	0.75 |.|	0.9 signed	0.9 |.|
30	1.000 S	1.000 S	1.000 S	0.994 S	1.000 S	1.000 S
31	1.000 S	0.977	1.000 S	0.958	1.000 S	0.999 S
32	1.000 S	0.987	1.000 S	0.957	1.000 S	0.885
33	1.000 S	0.964	1.000 S	0.954	1.000 S	1.000 S
34	1.000 S	1.000 S	1.000 S	0.964	1.000 S	1.000 S
35	1.000 S	0.944	1.000 S	0.967	1.000 S	0.825
36	1.000 S	0.977	1.000 S	0.993	1.000 S	0.960
37	1.000 S	0.931	1.000 S	0.958	1.000 S	0.964

Table C2. Columns are p_plus and arm. Registered (`RING_DIAL_ANALYSIS.txt`): p_plus 0.5 d -0.0275 (perm p 0.0070, 95% CI [-0.046, -0.009]), SOLVED 2/8 vs 8/8 (Fisher p 0.0070); 0.75 d -0.0320 (p 0.0002), 1/8 vs 8/8 (p 0.0014); 0.9 UNMEASURED, d -0.0459 (p 0.0256, below the 0.05 materiality floor), 4/8 vs 8/8 (p 0.0769).

C3. Three times the budget (Sec. 4.1; `RING_BUDGET_ANALYSIS.txt`; |.|, ring p_plus 0.5)

seed	900 epochs	300 epochs
40	1.000 SOLV (0.002)	0.957 STAL (0.150)
41	0.986 STAL (0.070)	0.982 STAL (0.098)
42	0.969 DESC (0.129)	0.946 STAL (0.252)
43	1.000 SOLV (0.002)	1.000 SOLV (0.003)
44	0.951 DESC (0.215)	0.895 DESC (0.301)
45	0.944 STAL (0.195)	0.998 SOLV (0.012)
46	1.000 SOLV (0.000)	1.000 SOLV (0.008)
47	0.989 DESC (0.056)	0.960 DESC (0.197)

Table C3. Registered PARTIAL: 3/8 vs 3/8 (Fisher p 1.00); the solved sets differ. No in-batch signed arm (the signed comparison, 8/8 at 300 epochs, is across batches and descriptive).

C4. Observation-step-free ring, from scratch and constructed (Secs. 4.1, 4.2; `WRAP_EXISTENCE_ANALYSIS.txt`)

seed	SO (signed)	AO (|.|)	AO_bal: frozen -> trained (max continuation loss)	SO_bal: frozen -> trained (max loss)
600	1.000 S	0.905 STAL	1.000 -> 1.000 S (0.041)	1.000 -> 1.000 S (0.019)
601	1.000 S	0.949 STAL	1.000 -> 1.000 S (0.030)	1.000 -> 1.000 S (0.042)

seed	SO (signed)	AO (|.|)	AO_bal: frozen -> trained (max continuation loss)	SO_bal: frozen -> trained (max loss)
602	1.000 S	0.853 STAL	1.000 -> 1.000 S (0.154)	1.000 -> 1.000 S (0.072)
603	1.000 S	0.951 STAL	1.000 -> 1.000 S (0.046)	1.000 -> 1.000 S (0.017)
604	1.000 S	0.907 STAL	1.000 -> 1.000 S (0.024)	1.000 -> 1.000 S (0.039)
605	1.000 S	0.930 STAL	1.000 -> 1.000 S (0.032)	1.000 -> 1.000 S (0.028)
606	1.000 S	0.950 STAL	1.000 -> 1.000 S (0.031)	1.000 -> 1.000 S (0.112)
607	1.000 S	0.922 STAL	1.000 -> 1.000 S (0.036)	1.000 -> 1.000 S (0.015)

Table C4. Registered: E3 SIGNED LEARNS BETTER (0.921 vs 1.000, d -0.079, perm p 0.0002; 0/8 vs 8/8, Fisher p 0.0002); E2 HELD (AO_bal 8/8 SOLVED, 0/8 LOST). The ill-conditioned constructions are in D3.

C5. The start manipulation (Sec. 4.3, Table 1; `MECHANISM_ANALYSIS.txt`)

seed	SO	AO	SO_same	SO_absmag	AO_absmag	AO_high	SO_high	AO_mid
800	1.000 S	0.907 STAL	1.000 S	1.000 S	0.865 STAL	1.000 S	1.000 S	0.968 S
801	1.000 S	0.894 STAL	1.000 S	1.000 S	0.922 STAL	1.000 S	1.000 S	1.000 S
802	1.000 S	0.910 STAL	1.000 S	1.000 S	0.861 STAL	1.000 S	1.000 S	0.917 STAL
803	1.000 S	0.966 STAL	1.000 S	1.000 S	0.860 STAL	1.000 S	1.000 S	0.921 STAL
804	1.000 S	0.987 STAL	1.000 S	0.866 STAL	0.933 STAL	1.000 S	1.000 S	0.976 DESC
805	1.000 S	0.888 STAL	1.000 S	1.000 S	0.857 STAL	1.000 S	1.000 S	0.890 STAL
806	1.000 S	0.880 STAL	1.000 S	1.000 S	0.889 STAL	1.000 S	1.000 S	0.928 STAL
807	1.000 S	0.925 STAL	1.000 S	1.000 S	0.945 STAL	1.000 S	1.000 S	0.951 STAL
mean	1.000	0.919	1.000	0.983	0.892	1.000	1.000	0.944

Table C5. SO_same = the default angles re-set in the balanced gauge (gauge control); SO_absmag / AO_absmag = the same-direction start; AO_mid = large start with sums below pi; AO_high / SO_high = large start with sums above pi. Registered (all n = 8): M5 AO_absmag vs SO_absmag SIGN COST AT THE |.| START (d -0.092 [-0.135, -0.045], perm p 0.0030; Fisher p 0.0014); M2 AO_high vs AO_absmag RESCUES (+0.109 [+0.085, +0.135], p 0.0002); M2b AO_high vs AO_mid (+0.056 [+0.030, +0.085], p 0.0014; Fisher p 0.0070); M6 AO_high vs SO_high NO EFFECT DETECTED; M1 SO_absmag vs SO_same UNMEASURED (-0.017); M3 gauge controls NO DIFFERENCE DETECTED; M4 AO vs SO SIGNED LEARNS BETTER (d -0.081, p 0.0002). Declared secondaries, end-of-training cancelling share: SO 0.86, SO_same 0.96, SO_absmag 0.89, AO 0.07, AO_absmag 0.04, AO_mid 0.05, AO_high 0.23, SO_high 0.34.

C6. Frozen move latents (Sec. 4.3; ACCOUNTS T2, `ACCOUNTS_ANALYSIS.txt`, seeds 1000-1007)

arm	start	move latents	mean accuracy	SOLVED	opposite-sign channels at the end, per seed
SO_same	default (balanced gauge)	free	1.000	8/8	1.00 1.00 0.98 0.98 1.00 1.00 1.00 1.00
SO_same_fz	same	frozen	0.991	7/8	0.51 0.73 0.22 0.38 0.45 0.77 0.68 0.54
SO_absmag	same-direction	free	0.985	7/8	1.00 1.00 1.00 0.41 1.00 1.00 0.19 1.00
SO_absmag_fz	same-direction	frozen	0.910	1/8	0.21 0.15 0.09 0.18 0.14 0.16 0.04 0.21

Table C6. Registered: F1 WORSE (1/8 vs 7/8, Fisher p 0.010; -0.075 [-0.130, -0.020], perm p 0.009); F2 NO EFFECT DETECTED; reading SUPPORTED. Per-seed accuracies are not printed in the analysis output; the run-level SOLVED counts are in C10.

C7. Torus, observation-step-free, from scratch and constructed (Secs. 4.1, 4.2; `GENERALITY_ANALYSIS.txt`)

seed	SO4	AO4	SO_bal (lr 1e-4)	AO_bal (lr 1e-4): peak loss, epoch of peak / of re-entry	AO_bal_hi (lr 1e-3)
700	1.000 S	0.874 STAL	1.000 S	1.000 S; 0.168, 29 / 153	1.000 S
701	1.000 S	0.707 STAL	1.000 S	1.000 S; 0.110, 40 / 121	1.000 S
702	1.000 S	0.946 DESC	1.000 S	1.000 S; 0.173, 19 / 204	0.999 S
703	1.000 S	0.879 STAL	1.000 S	1.000 S; 0.143, 52 / 149	0.950 STAL
704	1.000 S	0.892 STAL	1.000 S	1.000 S; 0.131, 13 / 119	0.960 DESC
705	1.000 S	0.907 STAL	1.000 S	1.000 S; 0.155, 13 / 148	0.992 S
706	1.000 S	0.877 STAL	1.000 S	1.000 S; 0.208, 36 / 164	1.000 S
707	1.000 S	0.960 DESC	1.000 S	1.000 S; 0.117, 23 / 139	1.000 S

Table C7. SO_bal's peak continuation losses 0.004-0.017. Registered: E3 SIGNED LEARNS BETTER (0.880 vs 1.000, d -0.120, perm p 0.0002; 0/8 vs 8/8); E2 NOT HELD by the LOST rule (6/8 LOST vs 0/8, Fisher on not-LOST p 0.0070), read by the rule fixed before launch as "perturbed and re-found" (8/8 SOLVED at the end). AO_bal_hi (descriptive): 6/8 SOLVED, 0.988.

C8. Torus start manipulation (Sec. 4.4; `TORUS_START_ANALYSIS.txt`)

seed	SO4	AO4	AO4_high	SO4_high	AO4_mid
1200	1.000 S	0.801 STAL	0.934 DESC	0.922 DESC	0.937 DESC
1201	1.000 S	0.838 STAL	0.794 STAL	1.000 S	0.908 STAL
1202	1.000 S	0.998 S	0.975 DESC	0.926 DESC	0.824 STAL
1203	1.000 S	0.899 STAL	0.931 DESC	0.905 DESC	0.906 DESC
1204	1.000 S	0.948 STAL	0.951 DESC	0.745 STAL	0.840 STAL
1205	1.000 S	0.875 STAL	0.876 STAL	0.927 DESC	0.949 DESC
1206	1.000 S	0.882 STAL	0.915 DESC	1.000 S	0.939 DESC
1207	1.000 S	0.927 STAL	0.828 STAL	0.928 DESC	0.990 DESC
mean	1.000	0.896	0.900	0.919	0.912

Table C8. Registered: TS1 NO EFFECT DETECTED (+0.005 [-0.060, +0.070], p 0.89); TS2 UNMEASURED; TS3 UNREADABLE (SO4_high 2/8 < 4); TS4 SIGNED LEARNS BETTER (1/8 vs 8/8, Fisher p 0.0014; -0.104 [-0.150, -0.060]). Strata (lag < 128 / >= 128 / wrap-only): SO4 1.000 / 1.000 / 1.000; AO4 0.939 / 0.730 / 0.632; AO4_high 0.949 / 0.777 / 0.534; SO4_high 0.970 / 0.716 / 0.627; AO4_mid 0.950 / 0.809 / 0.621. An untested account of why the past-pi start does not transfer (moved from the main text of draft 5): with 64 cells per axis and two inverse pairs per channel, a random large-angle code is far from any map. The start lever is a ring result, not a design rule.

C9. Moves +1/-1 vs +1/+2 (Sec. 4.5; `GAUGE_RESULTS.md`)

seed	SO_m	AO_m	SO_p	AO_p	RoPE_p
1300	1.000 S	0.904 STAL	0.987 STAL	0.987 DESC	0.642 STAL
1301	1.000 S	0.953 DESC	1.000 S	0.910 STAL	0.663 STAL
1302	1.000 S	0.957 STAL	1.000 S	0.904 STAL	0.655 STAL
1303	1.000 S	1.000 S	1.000 S	0.590 STAL	0.661 STAL
1304	1.000 S	0.922 STAL	0.935 STAL	0.768 STAL	0.654 STAL
1305	1.000 S	0.928 STAL	1.000 S	1.000 S	0.602 STAL
1306	1.000 S	0.880 STAL	0.654 DESC	0.908 STAL	0.660 STAL
1307	1.000 S	0.903 STAL	1.000 S	1.000 S	0.660 STAL

Table C9. _m = moves +1/-1, _p = +1/+2. Declared secondaries (revisit strata): on +1/-1, 89% of revisits are backtracks (138,800 of 155,456 on the eval stream; 87.5% in the pre-launch audit's sample; `GAUGE_RESULTS.md` carries a CORRECTED banner); on +1/+2 every revisit is a lap with gap >= 16.

cell	mean accuracy	SOLVED	accuracy 4T	loss AUC	backtrack	lap	gap < 16	gap >= 16
SO_m	1.0000	8/8	0.997	0.090	1.000	1.000	1.000	1.000
AO_m	0.9309	1/8	0.871	0.537	0.936	0.893	0.952	0.902
SO_p	0.9470	5/8	0.913	0.515	--	0.947	--	0.947
AO_p	0.8834	2/8	0.863	0.661	--	0.884	--	0.884

cell	mean accuracy	SOLVED	accuracy 4T	loss AUC	backtrack	lap	gap < 16	gap >= 16
RoPE_p	0.6494	0/8	0.468	1.341	--	0.651	--	0.651

Table C9b. The laps comparison quoted in Sec. 4.5 (|.| 0.893 vs 0.884) is a post hoc reading of these declared secondaries. Signed models on +1/+2 end with 0.92 of channels having both (wrapped) move angles of one sign (per seed 0.57-1.00), against 0.01 on +1/-1.

C10. SOLVED under four criteria (post hoc; `docs/audits/2026-10-15/reanalyses/solved_thresholds_out.md`)

Predictions and decision rules were written before the run (`PREDICTIONS.md` there); all 47 arms reproduce their registered SOLVED counts at the registered 0.05.

batch	contrast	loss < 0.02	loss < 0.05 (registered)	loss < 0.1	accuracy >= 0.99
SIGN_MATCHED torus	|.| vs signed	0/8 vs 8/8 (p 0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 7/8 (0.0014)
SIGN_MATCHED torus	softplus vs signed	0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)	1/8 vs 7/8 (0.0101)
RING_DIAL p 0.5	|.| vs signed	2/8 vs 8/8 (0.0070)	2/8 vs 8/8 (0.0070)	**4/8 vs 8/8 (0.0769)**	2/8 vs 8/8 (0.0070)
RING_DIAL p 0.75	|.| vs signed	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)	2/8 vs 8/8 (0.0070)	2/8 vs 8/8 (0.0070)
RING_DIAL p 0.9	|.| vs signed	4/8 vs 8/8 (0.0769)	4/8 vs 8/8 (0.0769)	4/8 vs 8/8 (0.0769)	4/8 vs 8/8 (0.0769)
WRAP_EXISTENCE ring	AO vs SO	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)
MECHANISM ring	AO vs SO	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	0/8 vs 8/8 (0.0002)
MECHANISM ring	AO_absmag vs SO_absmag	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)
MECHANISM ring	AO_high vs AO_absmag	8/8 vs 0/8 (0.0002)	8/8 vs 0/8 (0.0002)	8/8 vs 0/8 (0.0002)	8/8 vs 0/8 (0.0002)
MECHANISM ring	AO_high vs AO_mid	8/8 vs 1/8 (0.0014)	8/8 vs 2/8 (0.0070)	8/8 vs 3/8 (0.0256)	8/8 vs 1/8 (0.0014)
MECHANISM ring	AO_high vs SO_high	8/8 vs 8/8 (1.00)	8/8 vs 8/8 (1.00)	8/8 vs 8/8 (1.00)	8/8 vs 8/8 (1.00)
ACCOUNTS T2 ring	SO_absmag_fz vs SO_absmag	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)
ACCOUNTS T2 ring	SO_same_fz vs SO_same	7/8 vs 8/8 (1.00)	7/8 vs 8/8 (1.00)	7/8 vs 8/8 (1.00)	7/8 vs 8/8 (1.00)
GENERALITY torus	AO4 vs SO4	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)
GENERALITY ring	CoPE vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	0/8 vs 8/8 (0.0002)
TORUS_START torus	AO4 vs SO4				

batch	contrast	loss < 0.02	loss < 0.05 (registered)	loss < 0.1	accuracy >= 0.99
		0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)
TORUS_START torus	AO4_high vs AO4	0/8 vs 0/8 (1.00)	0/8 vs 1/8 (1.00)	0/8 vs 1/8 (1.00)	0/8 vs 1/8 (1.00)
ACCOUNTS T4	CoPE line vs signed line	0/16 vs 8/8 (< 1e-4)	1/16 vs 8/8 (< 1e-4)	8/16 vs 8/8 (0.0222)	7/16 vs 8/8 (0.0095)
ACCOUNTS T4	CoPE line vs CoPE ring	0/16 vs 0/16 (1.00)	1/16 vs 0/16 (1.00)	8/16 vs 2/16 (0.0538)	7/16 vs 2/16 (0.1134)

Table C10. Two-sided Fisher p in brackets. Within this table only the RING_DIAL p_plus 0.5 contrast changes its reading (at loss < 0.1). Outside it, ROBUST's torus SOLVED contrast does not survive loss < 0.1 (5/8 vs 8/8, Fisher p 0.20; a declared secondary of that batch, C13).

Where loss-based SOLVED and held-out accuracy disagree (reproduction audit, 2026-10-16). SOLVED is the train-mode (dropout on) final-5% loss. In ACCOUNTS T4, 5/16 CoPE line runs score 0.9997-1.000 held out (NLL <= 8e-4) with loss tails 0.062-0.089, so they count as unsolved, while the one SOLVED line run (s1104) scores 0.988; the CoPE ring arm has 2/16 at

> = 0.9999 and 0/16 SOLVED. At a held-out accuracy cut of 0.999 the line reads 5/16 vs 2/16 instead of 1/16 vs 0/16 (post hoc; no test run). Single runs show the same pattern: WRAP_COPE CoPEF_Hi s904 (0.9997, tail 0.055), CoPEF_LoS s902 (0.9992, 0.059), ROBUST ring lr x3 AO s1400 (1.000, 0.138), WRAP_EXISTENCE AO_wrap s604 (0.9998, 0.162); conversely MECHANISM AO_mid s800 is SOLVED at 0.968. SIGN_MATCHED Pos_r4 s7 is the softplus arm's only SOLVED run at tail 0.049 (knife-edge); that contrast fires on accuracy regardless.

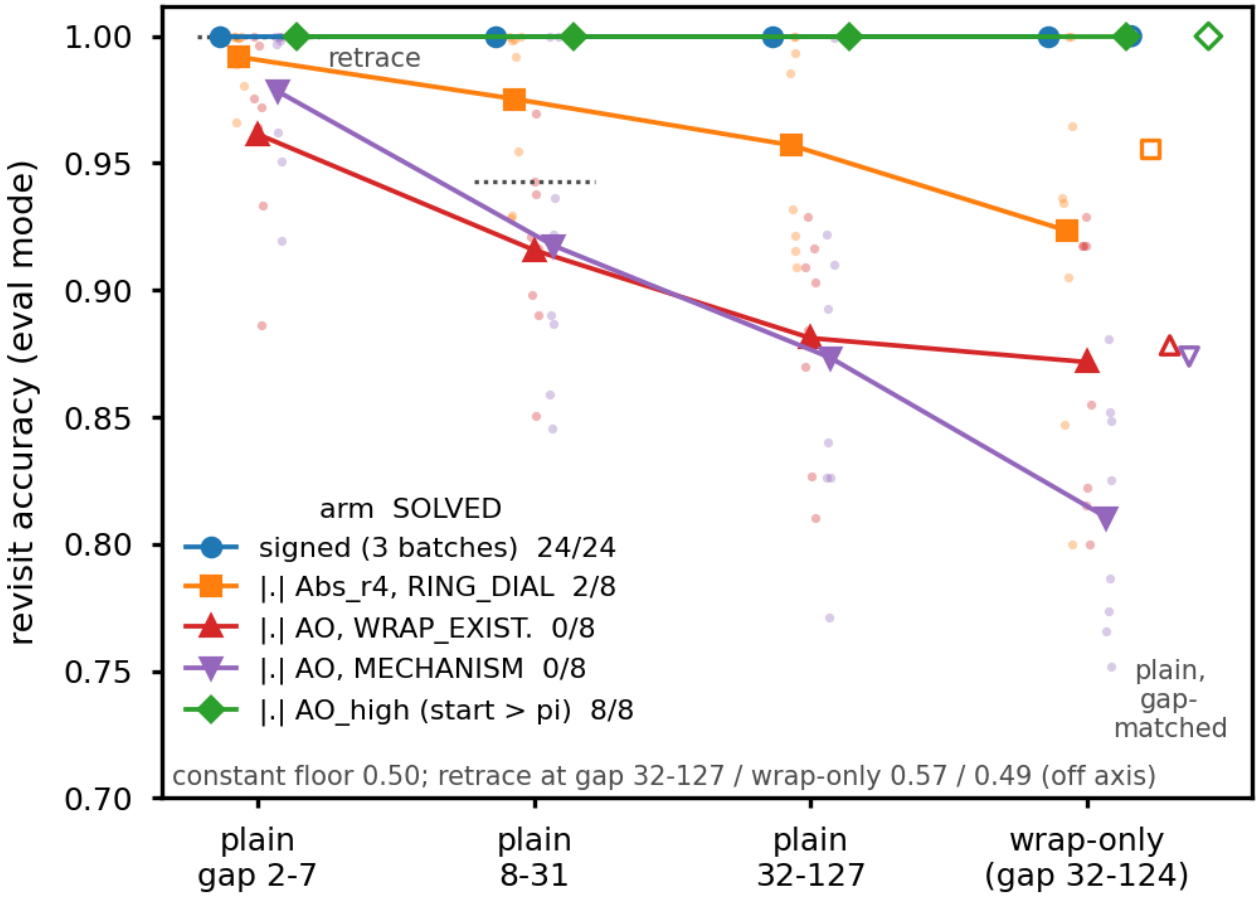
C11. Ring revisit strata and Fig. C1 (post hoc; `docs/paper/figs/ring_strata_out.txt`, `FIGS_NOTES.md`)

A revisit's *gap* is the number of moves since the last visit to the cell; a *wrap-only* revisit has every earlier visit at unwrapped displacement +-32k with k != 0. Stream: the registered ring eval stream (400 walks); overall accuracy reproduces every run's registered value (max |diff| 5e-5).

batch	arm	SOLVED	plain gap 2-7	plain 8-31	plain 32-127	wrap-only	plain, wrap-gap-matched	all
(stream)	n targets		12183	17420	5110	4134		38847
RING_DIAL	Signed_r4	8/8	1.000	1.000	1.000	1.000	1.000	1.000
RING_DIAL	Abs_r4	2/8	0.992 (0.97-1.00)	0.975 (0.93-1.00)	0.957 (0.91-1.00)	0.923 (0.80-1.00)	0.956	0.973
WRAP_EXISTENCE	SO	8/8	1.000	1.000	1.000	1.000	1.000	1.000
WRAP_EXISTENCE	AO	0/8	0.961 (0.89-1.00)	0.916 (0.85-0.97)	0.881 (0.81-0.93)	0.872 (0.80-0.93)	0.878	0.921

batch	arm	SOLVED	plain gap 2-7	plain 8-31	plain 32-127	wrap-only	plain, wrap-gap-matched	all
MECHANISM	SO	8/8	1.000	1.000	1.000	1.000	1.000	1.000
MECHANISM	AO	0/8	0.978 (0.92-1.00)	0.918 (0.85-1.00)	0.873 (0.77-1.00)	0.811 (0.75-0.88)	0.874	0.919
MECHANISM	AO_high	8/8	1.000	1.000	1.000	1.000	1.000	1.000

Table C11 (Fig. C1, `docs/paper/figs/fig_ring_strata.png`). Means over runs, range in brackets. |.| models lose accuracy with the gap in every batch (gap 2-7 to 32-127: -0.035 for Abs_r4, sign-flip p 0.031; -0.080 and -0.105 for the two AO batches, 8/8 runs lower, p 0.0078 each). The wrap-only deficit beyond gap-matched plain revisits varies between batches: -0.063 (MECHANISM AO, p 0.031), -0.007 (WRAP_EXISTENCE AO, p 0.88), -0.032 (Abs_r4, p 0.31). Signed models are at 1.000 in every stratum. Uncorrected for 3 batches x 2 contrasts; accuracy tracks loss, so the strata locate errors, not a mechanism. The constant predictor scores 0.49-0.50 in every stratum.



C12. Loose ends of the ring mechanism (post hoc; moved from the main text of draft 4)

Rules for the wall check and the channel-by-channel flow were fixed before the run (`docs/audits/2026-10-15/reanalyses/PREDICTIONS.md`).

- **The |.| start is small in most channels, not all.** Per-channel initial sums span 0.01-6.58 rad; 5% of channels start past pi (`docs/paper/figs/fig_mechanism_out.txt`; `docs/audits/2026-10-14/theory_review/mech_channels_out.txt`). Of the channels whose step does not collapse to zero, the 20 (over 8 runs) that start with sums between pi and 3 pi / 2 end with none cancelling (`mech_bins_out.txt`): being past pi is not sufficient per channel.
- **Rescued models use the wrap sparingly.** From past pi, |.| models solve with 23% of channels cancelling (signed 34%; 7% at the start by chance; `MECHANISM_RESULTS.md` secondaries). Capture falls gently with distance from 2 pi: 0.34 within 0.25 rad, 0.08 at 2-3 rad (`mech_capture_out.txt`).
- **The one-channel flow does not predict which channels cancel.** Gradient flow on the exact one-channel retrieval potential at a fresh model's attention scale (appendix E, Remark R) reproduces neither the signed escape (12% of channels cross zero) nor a sharp past-pi rescue (only channels within ~0.5 rad of 2 pi are captured: 28% of a U(1.7, 4.5) start). Run from each AO_high channel's own start (505 channels over 8 runs; `capture_by_channel_out.txt`) it predicts 0.329 cancelling against 0.230 observed; agreement with the trained models is barely above chance (Cohen's kappa +0.086; per run -0.26 to +0.51); it captures 95% of the channels starting within 0.5 rad of 2 pi, where the models cancel 28%; 59% of the models' cancelling channels are ones it does not capture. For SO_high, kappa +0.198. Whether the rescue is collective is untested; what the other 77% of channels do is unknown.
- **The wall.** 44% of channels in |.| models from the same-direction start end with the forward-move angle within 0.05 rad of zero, against 17% for signed models from that start and 0.2% and 1.4% for |.| from the large starts (`fig_mechanism_out.txt`). Per move step (`collapse_wall_out.md`): collapse is common to every |.| arm from a small start (ring 0.235-0.424, torus 0.601-0.695, softplus 0.624 included) and smaller in signed controls from the default start (ring 0.104-0.157, torus 0.343-0.402). It is partly inherited: of the |.| steps that start within 0.05 rad of zero, 77-81% end there (signed 29-35%). Attention reads the collapsed |.| channels (usage ratio collapsed / not 0.39-1.10). At the end of training the loss gradient points into the wall for 51-57% of the collapsed steps in the AO arms (eval mode, ring eval stream), which the pre-stated rule reads as MIXED, not as P4. Torus gradients were not computed.

batch	arm	collapsed move steps, mean (run range)
SIGN_MATCHED torus	Signed_r4 / Abs_r4 / Pos_r4	0.343 (0.28-0.41) / 0.695 (0.62-0.88) / 0.624 (0.54-0.71)
RING_DIAL p 0.5 / 0.75 / 0.9	Signed_r4	0.104 / 0.107 / 0.156
RING_DIAL p 0.5 / 0.75 / 0.9	Abs_r4	0.302 / 0.270 / 0.235
	Abs_r4	0.366

batch	arm	collapsed move steps, mean (run range)
RING_BUDGET (900 epochs)		
WRAP_EXISTENCE	SO / AO	0.157 / 0.337
MECHANISM	SO / SO_absmag / SO_high	0.145 / 0.171 / 0.001
MECHANISM	AO / AO_absmag / AO_mid / AO_high	0.368 / 0.424 / 0.010 / 0.001
GENERALITY torus	SO4 / AO4	0.357 / 0.660
TORUS_START torus	SO4 / AO4 / SO4_high / AO4_high / AO4_mid	0.402 / 0.601 / 0.062 / 0.061 / 0.127

Table C12 (`collapse_wall_out.md (a)`). Collapsed = |a| <= 0.05 rad, over (move, channel) entries.

- **Start size.** A start nearer the wrap also moves the angles 13-22x faster per Adam step; the sub-pi control (AO_mid, ~5x) separates sum from size only partly (`MECHANISM_RESULTS.md`, M2b).

C13. Recipe and scale (Sec. 4.1; `ROBUST_RESULTS.md`, `ROBUST_ANALYSIS.txt`; registered, seeds 1400-1407)

Observation-step-free SignedAO / AbsAO. Ring: `train_gauge.py --deltas=1,-1`, 32 cells, T = 128, 300 epochs, four recipes. Torus: GENERALITY's recipe at 2 layers, d_model 256 (1.59M parameters), T = 1024, 900 epochs. Reading: PARTLY UNMEASURED.

Registered decisions (AO vs SO; Fisher on SOLVED, permutation on accuracy) | cell | recipe | AO acc, SOLVED | SO acc, SOLVED | d [95% CI] | perm p | Fisher p | label | |---|---|---|---|---|---|---|---| | ring def | lr 1e-3, wd 0.05 | 0.911, 1/8 | 1.000, 8/8 | -0.089 [-0.125, -0.055] | 0.0002 | 0.0014 | **SIGNED LEARNS BETTER** | | ring wd0 | lr 1e-3, wd 0 | 0.873, 0/8 | 1.000, 8/8 | -0.127 [-0.200, -0.075] | 0.0002 | 0.0002 | **SIGNED LEARNS BETTER** | | ring lrx3 | lr 3e-3, wd 0.05 | 0.987, 4/8 | 1.000, 8/8 | -0.013 [-0.025, -0.005] | 0.026 | 0.077 | **UNMEASURED** | | ring lrd3 | lr 3.3e-4, wd 0.05 | 0.841, 0/8 | 0.950, 6/8 | -0.109 [-0.215, +0.010] | 0.089 | 0.0070 | **SIGNED LEARNS BETTER** | | torus d256L2 | 2 layers, d 256 | 0.967, 2/8 | 0.999, 8/8 | -0.032 [-0.055, -0.015] | 0.0005 | 0.0070 | **SIGNED LEARNS BETTER** |

Gates passed in every cell (SO >= 4/8). lrx3 is UNMEASURED, not NO SIGN COST: SOLVED differs by 4 (Fisher p 0.077) and the accuracy gap (perm p 0.026) is smaller than the registered 0.05 materiality. lrd3's accuracy contrast is not detected (perm p 0.089, CI [-0.215, +0.010]); its label rests on SOLVED. Across recipes at matched seeds (descriptive) |.| solves 0/8 (lr /3), 1/8 (default), 4/8 (lr x3): the comparison is at the signed arm's default rate, and the lr cells also change the effective decay (lr x wd) and the cosine floor (ROBUST Amendment 1). Floors (post hoc): torus on this batch's walks blank 0.509, n-gram 0.578, retrace 0.844, every run of both arms above retrace (lowest |.| 0.910); torus strata (lag < 128 /

> = 128 / wrap-only) SO 1.000 / 0.999 / 0.992, AO 0.989 / 0.911 / 0.799. The SOLVED-at-loss-0.1 column below is a declared secondary (not post hoc).

Declared secondaries:

cell	arm	acc 4T (ring 512) / 2T (torus 2048)	loss AUC	SOLVED at loss 0.1	wrap-cancel	opposite-sign
ring def	SO / AO	0.996 / 0.773	0.117 / 0.607	8/8 / 1/8	0.85 / 0.07	0.99 / 0.04
ring wd0	SO / AO	0.978 / 0.758	0.150 / 0.796	8/8 / 1/8	0.43 / 0.11	0.86 / 0.05
ring lrx3	SO / AO	1.000 / 0.934	0.077 / 0.453	8/8 / 4/8	1.00 / 0.13	1.00 / 0.07
ring lrd3	SO / AO	0.911 / 0.747	0.294 / 0.825	7/8 / 0/8	0.69 / 0.04	0.79 / 0.03
torus	SO / AO	0.983 / 0.888	0.070 / 0.319	8/8 / 5/8	1.00 / 0.11	1.00 / 0.06

Per seed (accuracy at the training length, class):

```
ring  seed | def:SO | def:AO | wd0:SO | wd0:AO | lrx3:SO | lrx3:AO | lrd3:SO | lrd3:AO
1400 | 1.0000 SOLV | 0.8230 STAL | 1.0000 SOLV | 0.9078 DESC | 1.0000 SOLV | 1.0000 DESC |
1.0000 SOLV | 0.8893 STAL
1401 | 1.0000 SOLV | 0.9155 STAL | 1.0000 SOLV | 0.6239 STAL | 1.0000 SOLV | 0.9596 STAL |
0.6194 STAL | 0.6221 STAL
1402 | 1.0000 SOLV | 0.9237 STAL | 1.0000 SOLV | 0.8878 STAL | 1.0000 SOLV | 0.9702 STAL |
1.0000 SOLV | 0.8319 STAL
1403 | 1.0000 SOLV | 0.9136 STAL | 1.0000 SOLV | 0.8800 STAL | 1.0000 SOLV | 0.9995 SOLV |
1.0000 SOLV | 0.9167 STAL
1404 | 1.0000 SOLV | 0.9118 STAL | 1.0000 SOLV | 0.9119 STAL | 1.0000 SOLV | 1.0000 SOLV |
1.0000 SOLV | 0.8313 STAL
1405 | 1.0000 SOLV | 0.8791 STAL | 1.0000 SOLV | 0.8858 STAL | 1.0000 SOLV | 0.9650 STAL |
1.0000 SOLV | 0.8526 STAL
1406 | 1.0000 SOLV | 0.9211 DESC | 1.0000 SOLV | 0.9110 STAL | 1.0000 SOLV | 1.0000 SOLV |
0.9794 STAL | 0.9062 STAL
1407 | 1.0000 SOLV | 0.9997 SOLV | 1.0000 SOLV | 0.9735 DESC | 1.0000 SOLV | 0.9998 SOLV |
1.0000 SOLV | 0.8801 STAL
torus 1400 SO 1.0000 SOLV | AO 0.9941 DESC     1404 SO 1.0000 SOLV | AO 0.9935 SOLV
      1401 SO 1.0000 SOLV | AO 0.9981 SOLV     1405 SO 1.0000 SOLV | AO 0.9706 DESC
      1402 SO 0.9941 SOLV | AO 0.9705 DESC     1406 SO 1.0000 SOLV | AO 0.9098 DESC
      1403 SO 1.0000 SOLV | AO 0.9454 DESC     1407 SO 1.0000 SOLV | AO 0.9544 DESC
```

D. Constructions

D1. Per channel (B0, P1.0)

For a signed step Delta and omega != 0, Delta' = Delta mod (2 pi / |omega|) lies in [0, 2 pi / |omega|) and omega Delta' = omega Delta (mod 2 pi) for either sign of omega. The wrap is taken in step space: the earlier form remainder(omega Delta, 2 pi) / omega is wrong for omega < 0, and trained omega does cross zero (pilot -0.012; RING_DIAL seeds 31-33) (`WRAP_EXISTENCE_PREREG.md` A2).

D2. The balanced construction (P1(a); the registered primary)

Source: `model_wrap_exist.wrap_construct(mode="balanced")`, `WRAP_EXISTENCE_PREREG.md` Amendment 1 (A1), on the torus `build_torus_exist.py`. Given a solved signed observation-

step-free model with m stepping move tokens (m = 2 ring, 4 torus; r = 4): 1. Re-solve W_in so that the move latents are c e_1, ..., c e_m (c = the source's mean move-latent norm). W_in acts only on the move embeddings (every other step is 0), so this is a free gauge. 2. Per channel set omega' = max(|omega|, 1) and the steps to remainder(omega Delta, 2 pi) / omega' (angles in [0, 2 pi) for either sign of omega; the code's form). 3. Set W_out[:, j] = step_j / c for the m move columns and 0 elsewhere. Every other weight is copied.

The signed control SO_bal is built by the same function at the same W_in and omega' with the *unwrapped* signed angle (steps omega Delta / omega'), so the two differ only in the wrap (same RNG consumption, same data stream). On the torus m = 4 = r, exactly at P1(a)'s limit.

Exactness (E1, registered). Ring: frozen accuracy = source on 8/8 seeds; max angle error 1.2e-6 rad (AO_bal), 6.1e-7 (SO_bal); max logit difference 5.1e-4 at T = 128, 1.1e-2 at T = 512 (`WRAP_EXISTENCE_ANALYSIS.txt`). Torus: frozen = source on 8/8; max angle error 1.4e-6 rad (AO_bal), 5.8e-7 (SO_bal) (`GENERALITY_ANALYSIS.txt`).

Weight scale. Ring: max |W_out| 0.31-0.38 in the signed sources, 1.07-1.70 built (AO_bal); |W_in| 0.15-0.21 -> 0.19-0.33; max step 6.28 (one turn) (`WRAP_EXISTENCE_ANALYSIS.txt`): within ~5x. Torus: max |W_out| 0.48-0.58 source -> 1.23-1.55 built (AO_bal), max |omega| 0.68-0.96 -> 1.00 (`GENERALITY_ANALYSIS.txt`).

D3. Why the naive construction fails as a test

Keeping the source's W_in and solving W_out (min-norm, `AO_wrap`) is exact (frozen accuracy 1.000 on 8/8, angle error 1.2e-5 rad) but ill-conditioned: in a solved signed ring model the two move latents are nearly antiparallel (cosine -0.987 to -0.998 over seeds 600-607), so W_out must be huge -- max |W_out| 653-16,320 against 0.31-0.38 signed, steps up to 9,333 (`WRAP_EXISTENCE_ANALYSIS.txt`). Under Adam's per-entry steps (rule 15: a parameterisation change is an optimiser change) the continuation loses the solution in the warm-up: LOST on 8/8 (max loss 0.6-2.0; Fisher on not-LOST p 0.0002 against SO_cont) and then re-finds it (7/8 SOLVED). A per-channel gauge version (`AO_wrapg`, max |W_out| 4.62 at seed 600) also loses it (8/8 LOST, 8/8 re-solved). By the reading fixed before launch, NOT HELD under an ill-conditioned construction says nothing about the class. The pilot reading of this as an intrinsic "path-length fragility" was confounded and is withdrawn (`WRAP_EXISTENCE_RESULTS.md`).

D4. The full |.| arms (observation steps present; ring audit M1)

To show that the *task* (not a given signed table) is representable by `Abs_r4`, the ring-dial audit chose W_in to project onto two directions orthogonal to all 17 observation embeddings and solved W_out per channel (`docs/audits/2026-10-12/ring_dial_audit.md` M1; `wrap_construction_t2.py`). In the real modules: |.| steps >= 0 (min 5e-11), observation-token |Delta| <= 1.7e-6, cos / sin equal to the signed code's to 6.3e-4 over a 512-step walk, circular mismatch <= 4.6e-4 on 2,000 same-cell revisit pairs. **Softplus (P1(f), appendix E; new in draft 6).** Softplus cannot reach 0, but observation steps need only be 0 mod 2 pi: with W_in annihilating every observation embedding, each observation steps softplus(0) = ln 2 in every channel, and omega'_c = 2 pi / ln 2 makes that exactly one turn; the move columns are set to softplus^{-1}((remainder(Theta, 2 pi) + 2 pi) /

omega'). Built in the real `MapFormerWM_Pos_r4` from the solved MECHANISM ring model SO_s800 (`docs/paper/review_draft5/theory_check_extra.py` (2), CPU): min step 0.693 > 0, angle error mod 2 pi 8.4e-15 rad over all 19 tokens, logits equal to the source's to 2.8e-11 (scale 21.6). On the torus m = 4 = r, so the same construction applies (not built). Existence under training (frozen and trainable continuation) was not tested.

D5. Continuations and the LOST rule

A constructed model is trained further on a fresh walk stream and compared with its signed control at the same seed. **LOST** (registered; `WRAP_EXISTENCE_PREREG.md` A4, `GENERALITY_PREREG.md`): the run's max per-epoch training loss during the continuation exceeds 0.3, or exceeds 10x its control's max at that seed and 0.05. **HELD** = SOLVED on >= n - 1 seeds AND mean accuracy >= control - 0.01 AND LOST on <= 1 seed. **NOT HELD** = Fisher (SOLVED) p < .05 with the control ahead, OR sign-flip p < .05 with d <= -0.05, OR Fisher (not LOST) p < .05 with more LOST in the constructed arm. Otherwise PARTIAL.

- **Ring** (lr 1e-3, 300 epochs, re-warmed schedule, fresh optimiser state): AO_bal HELD (8/8 SOLVED, 0/8 LOST; max per-epoch loss 0.024-0.154 vs the control's 0.015-0.112; Table C4).
- **Torus** (lr 1e-4 primary): at lr 1e-4 the control's losses are small, so LOST reduces to "left the solved region at some epoch"; the reading fixed before launch was that NOT HELD by LOST with SOLVED >= n - 1 at the end means "perturbed and re-found, not held in place" (`GENERALITY_PREREG.md` Amendment 1 A3). Why lr 1e-4: in the pilot (seed 690), re-warming to lr 1e-3 lifted the *signed* control itself from loss ~0 to ~0.1, so at 1e-3 the continuation cannot separate the arms; 1e-4 is about the learning rate at which the source solution was reached (`GENERALITY_PREREG.md`). At lr 1e-3 (descriptive): AO_bal_hi 6/8 SOLVED (0.988), max loss 0.46-0.63, SO_bal_hi 8/8 (max loss 0.15-0.21) (Table C7).
- Ring and torus E2 labels are not directly comparable (continuation lr 1e-3 vs 1e-4).

D6. What the held model keeps (ACCOUNTS T1, registered)

Weighted by usage (query x key norm per channel), the held ring AO_bal keeps 0.99 -> 0.98 of its cancelling code; the unweighted share falls 0.87 -> 0.18 because unused channels leave the code (`ACCOUNTS_RESULTS.md`). T1's wd 0.05 runs are bit-identical retrains of WRAP_EXISTENCE's AO_bal / SO_bal (reproduction audit 2026-10-16, integrity note 2), so T1 is a second readout of the same runs, not a replication of E2 HELD; its wd 0 arms are new runs. Weight decay 0 vs 0.05 changes nothing detected (D1 -0.003, NO EFFECT DETECTED). WRAP_EXISTENCE's secondary "87% -> 18% drift" and the reading "not an attractor" are withdrawn. On the torus the unweighted share goes 1.00 at construction -> 0.69 after training (GENERALITY secondary; not usage-weighted). The frozen perturbation probe (1% multiplicative noise on omega drops the wrapped ring solution to 0.802, the signed one stays 0.998) is a property of frozen models at a matched gauge, not of training (appendix E, P2).

F. Dyck-2 at matched depth

Source: `DYCK_MDEPTH_RESULTS.md`, `DYCK_MDEPTH_PREREG.md`, driver `run_dyck_mdepth.sh`, trainer `train_dyck.py`.

Task and metric. Dyck-2 (MapFormer v4 App. B.4), trained and scored at length L = 32 and nesting depth D = 12. Primary metric A2f: Hewitt's distance-averaged closing accuracy over prefixes where the sampler can emit a closer (the CE-optimal predictor scores 1.000; chance 0.500); best map-free floor 0.594 (appendix B).

Models and recipe. 2 heads, d_model 128; 1-4 layers; path-integrated arms MapWM and MapPoPE at rank 2 (`--rank 2`), index arms RoPE and PoPE at base 10000 (and base 32 in the 4-layer 3x cell); omega base 32 (the training context). AdamW lr 1e-4, weight decay 0.01, batch 128, 5% linear warm-up then cosine to 10% of the peak, dropout 0.1, next-token cross-entropy on every position; 560,000 training sequences (1x) or 1,680,000 (3x). n = 8 seeds (0-7), paired.

Results (registered). Position main effect = ((MapWM - RoPE) + (MapPoPE - PoPE)) / 2, A2f at L32 D12: - Depth ladder at 1x budget (declared secondary S1): +0.353 / +0.130 / +0.045 / +0.024 at 1 / 2 / 3 / 4 layers (8/8 seeds positive each, sign-flip p 0.0078; MDE 0.003 / 0.025 / 0.009 / 0.007). - Primary, 4 layers at 3x budget: every arm at ceiling (index 0.997-0.998, path 1.000); effect +0.002: CLOSES. The earlier ladder's +0.168 at 4 layers was depth extrapolation (trained at D = 4). - Mixture training (D drawn from 4..12, 4 layers): +0.110 at L32 D12 (8/8). - Index base 32 vs 10000 at 4 layers: within MDE (+0.001 / -0.001). The 1x budget limits the index arms (3x - 1x +0.021).

So one layer of path integration is worth about three layers of index attention on Dyck-2 at matched depth (at 3 layers the gap is +0.045; at 4, +0.024).

Protocol. The batch (2026-09-26) predates the independent code checks. Disclosed before registration: four n = 1 smoke runs with non-default flags at seed 0, whose training losses were seen (no A2 / A2f computed); the batch retrained seed 0 from scratch and nothing from the smoke runs enters it. The in-batch reproduction of two stored ladder runs was weight-exact.

G. Navigation in words

Source: `TEXTWORLD_RESULTS.md` (its AUDIT and CORRECTED blocks first), driver `run_textworld.sh`, trainer `train_textworld.py`.

Task. The 64 x 64 torus walk rendered as English: 58 words, three synonyms per direction, movement-free verbs, fillers and asides; T = 1024 words (~132 steps); held-out map; predict the observation word at revisits. Floors: best constant 0.505, reversal-copy rule 0.597 (appendix B).

Recipe. As the torus (A3): AdamW lr 1e-3, batch 16, 98 batches per epoch, 900 epochs, cosine; path MapWM r = 4 at one layer vs index RoPE at one and two layers; 200 evaluation walks.

Results. Registered (8 seeds per arm): path 0.969 +/- 0.054 (7/8 SOLVED) vs index one layer 0.505 (0/8) and two layers 0.772 (0/8); path - index one layer +0.464 (exact permutation p 0.0002; Fisher p 0.0014). **Seeds 0 and 1 are the pilot** (byte-identical to `runs/textworld_pilot`), so the numbers to quote are the six fresh seeds: path 0.979 +/- 0.048 (6/6 SOLVED) vs index 0.505 (0/6), +0.474, permutation and Fisher p 0.0022; index two layers 0.793. Dropout-scale re-score (AUDIT 2026-10-04, all 24 runs): path 0.969 -> 0.993, index 0.505 -> 0.506 and 0.772 -> 0.772; the verdict only strengthens. Declared secondaries: with the common component of the four direction steps removed, north + south and west + east cancel to 0.002-0.012 on all 8 seeds; 4/8 seeds also run a per-step clock (31-38 of 64 phase channels drift between visits to the same cell; CORRECTED block). The registered step-table branch (B) did not fire (4/8).

Scope. Scripted grammar; a direction word never appears outside a movement clause (context-free steps).

H. Faithful CARoPE and CoPE

Sources: `model_mono_baselines.py` (docstring), `GENERALITY_RESULTS.md`, `WRAP_COPE_RESULTS.md`, `ACCOUNTS_RESULTS.md` T4, `docs/audits/2026-10-14/cope_diagnosis/COPE_DIAGNOSIS.md` (post hoc), appendix E P6 and P7.

CARoPE (`CARoPEF`). Per head h and token t a scalar rate f_h(x_t) = 1 / (softplus(x_t W)_h + 1) in (0, 1) replaces RoPE's base frequency, so channel i advances by f_h(x_t)^i per token. Initialised as index RoPE (W = 0, a per-head bias with 1 / (softplus(b) + 1) = 10000^(-2 / d_head)); the bias is ours (without it the start is RoPE with base ~2e7). Every token, observations included, advances the phase; the layer is MapWM's, so only the phase differs. *P6 (proved, appendix E):* the shortest return advances each channel by a value in (0, 4] < 2 pi, so no channel reads every return at phase 0; a channel that reads the short returns is blind over the excursion's range. Registered (GENERALITY): torus 0.667 (0/8), WORSE THAN INDEX (0.722, -0.055, p 0.040); ring 0.741 vs index 0.730, UNMEASURED (+0.011, p 0.39).

CoPE (`CoPEF`). Golovneva et al.'s Listing 1, verbatim in torch: per layer one module shared by the heads; gates g_ij = sigmoid(q_i . k_j / sqrt(d_h)); count p_ij = sum_{k=j}^{i} g_ik, clamped to npos_max - 1 = 2047; position logit by linear interpolation of q_i . e[floor p] and q_i . e[ceil p], table e zeros at initialisation; added to the attention logits; no rotary position. *Representability (constructed, GENERALITY audit):* gates 17/16 and 15/16 per (move, observation) pair make p_ij mod 2 a function of net displacement, read by a period-2 table (1.000 revisit accuracy). *P7 (appendix E, proved under its setup):* the exact solutions form a lattice (period P = 2 or 3; alpha an odd multiple of P/32, i.e. move-gate difference an odd multiple of P/16).

Prior measurement. MapFormer v4 already reports CoPE failing its 1D and 2D navigation tasks in distribution (Table 2: 1 layer 0.63 +/- 0.21 in 1D and 0.65 in 2D, 4 layers 0.88 and 0.76, MapWM-r2 1.00) and attributes it to the gate's sign ("cannot represent an action and its inverse", their App.). So "CoPE fails these maps" is a replication; what is new here is that CoPE can represent ring cancellation and still fails, the line control, and the CARoPE results.

Results. | batch | task | CoPE | SOLVED | comparison | |---|---|---|---|---| | GENERALITY (700-707) | ring | 0.887 | 0/8 | signed 1.000, 8/8 (-0.113, p 0.0002); index 0.730 (+0.157, p 0.0002) | | WRAP_COPE (900-907) | ring, table at zeros (the paper's) / sinusoid past pi / below pi / slope-matched | 0.840 / 0.913 / 0.867 / 0.943 | 1/8, 2/8, 1/8, 0/8 | all contrasts UNMEASURED (bimodal seeds) | | ACCOUNTS T4 (1100-1115) | line (256 cells) / ring | 0.914 / 0.873 | 1/16 / 0/16 | line: signed 1.000 8/8, index 0.772 0/8; L1 line vs ring NO EFFECT DETECTED; L2 SIGNED LEARNS BETTER |

Per seed (ACCOUNTS T4): CoPE line 0.836 0.868 0.867 0.824 0.988 1.000 0.848 0.620 1.000 0.886 0.999 1.000 0.894 1.000 0.998 1.000; CoPE ring 0.850 0.872 0.799 0.874 0.834 1.000 0.797 0.666 0.915 0.861 0.957 0.896 0.823 0.974 0.848 1.000. Several CoPE seeds reach 0.99-1.00 accuracy without meeting the loss criterion; at loss < 0.1 half the line runs pass (8/16 vs 2/16 on the ring, Fisher p 0.054; post hoc, Table C10), and at a held-out accuracy cut of 0.999 the line reads 5/16 vs 2/16 (post hoc; C10 note). The loss-based SOLVED count is fragile for CoPE, so the line-vs-ring comparison is unmeasured, not null. The "1 of 32 runs at the paper's initialisation" of Sec. 4.6 pools GENERALITY (0/8), WRAP_COPE zeros (1/8) and ACCOUNTS T4 ring (0/16), post hoc.

Diagnosis (post hoc, eval-only, 40 ring runs; predictions written before any checkpoint was opened). Runs fall into three regimes: NONE (20/40 runs have a query type with no locked count; identity-dependent observation gates), OFF (11 runs: the count locks, local revisits are solved, but the move-gate difference is off the lattice, so wrap-only revisits sit near the floor, mean 0.55), LATTICE (9 runs, all 4 SOLVED runs among them). In the OFF runs an eval-only snap of the move gates to the nearest lattice point raises wrap-only accuracy 0.63 -> 0.90 and the straight path to the lattice rises in loss before it falls (11/11). The registered line test then showed the wrap is not CoPE's main obstacle (T4). The precision a count needs (on the line, a margin of at most P / (D_max + 1) ~ 0.016; appendix E, P7) is a plausible account, not a tested one; the near-wrap table start does not measurably help (WRAP_COPE, unmeasured).

I. The toy (exploratory, not registered)

Source: `MECHANISM_PREREG.md` (exploratory section and Amendment 1, A1), `ACCOUNTS_RESULTS.md` T3, `runs/toy_wrap/summary.txt`, `runs/accounts/t3/summary.txt`. A 64-channel retrieval model with no transformer and no observation tokens: two move angles per channel (rank-4 latents, signed or |.|, MapWM's frequency ladder), softmax retrieval of earlier moves at the same ring cell by sum_c w_c cos(theta_c(t) - theta_c(t')). SOLVED = retrieval accuracy >= 0.99.

start (move angles)	signed	|.|
default	6/12 (median 0.946)	0/12 (0.516)
|default| (the |.| start)	0/8 (0.406)	0/8 (0.385)
large, sums below pi, U(0.9, 1.5)	0/8	0/8
large, sums above pi, U(1.7, 4.5)	8/8	11/11

It reproduces the |.| failure without a transformer, but its signed model fails often from the default start and never escapes the |.| start (0/8), unlike the transformer (7/8). An observation-retrieval target (ACCOUNTS T3, seeds 10-17) does not open the escape

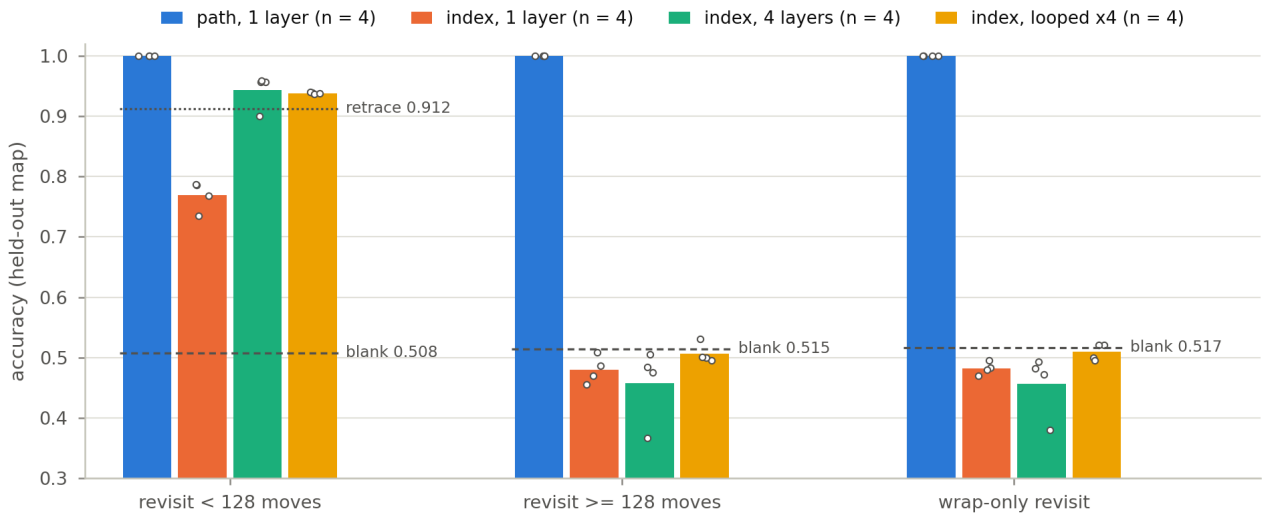
either (signed from the |.| start 0/8 MAP SOLVED; opposite-sign channels 0.27 vs 0.23 with the cell target). The toy and the transformer differ for an unidentified reason; the toy's stronger "the cost is the start, not the constraint" did not transfer and is not claimed.

J. The clock side is not established

A barrier step might learn a content *clock* faster than a signed one. On k-back recency (k_max 128, T = 1024, 300 epochs, one layer) it trends that way and is unmeasured twice (registered): loss AUC (mean training loss) |.| - signed -0.197 (exact sign-flip p 0.053, faster on 12/16 seeds; `CLOCK_SPEED_RESULTS.md`), then -0.106 on 16 fresh seeds (p 0.19, 10/16; `CLOCK_SPEED_REP_RESULTS.md`). Pooled post hoc -0.151 (p 0.018), pre-committed not to upgrade the verdict. On accuracy (`CLOCK_SIDE_RESULTS.md`) every barrier-vs-signed contrast is UNMEASURED (|.| +0.085, CARoPE +0.169, softplus +0.084; perm p 0.16-0.53) and no run converged (0/40 SOLVED); index RoPE fails the task (-0.450, p 0.0011). We do not claim a double dissociation. A consistent but untested account: signed steps contain every non-negative path, so a clock costs them a delay (moving minority signs through zero), while a map costs |.| a block.

K. Where a map is needed, one path layer beats index attention at the depths we tried

On the torus at T = 1024 (FACTORIAL, n = 4 per cell, seeds 500-503), one signed path layer scores 1.000 (4/4 SOLVED) against index RoPE at one layer 0.719, four layers 0.860 and a block looped four times 0.864 (0/4 each, all stalled within 900 epochs; complete separations, p 0.029; floors: blank 0.507, n-gram 0.576, computed on another stream of the same generator, appendix B). Depth and looping lift index accuracy only on revisits within 128 moves (0.74-0.79 -> 0.90-0.96); long-gap (>= 128 moves) and wrap-only revisits stay at 0.37-0.53, near or below the blank floor (Fig. K1, `docs/paper/figs/fig_factorial_strata.png`; its dashed line is the mean of the per-stratum best-constant floors on FACTORIAL's own stream, 0.513, overall 0.509; the 0.507 / 0.576 above are another stream of the same generator). Told in English words (appendix G; pilot seeds excluded): path 0.979 vs index 0.505, +0.474 (6/6 vs 0/6 SOLVED, p 0.0022); the eight-seed registered numbers are 0.969 vs 0.505 (p 0.0002), and with the dropout-scale re-score the path arm reads 0.993.



This is a learnability measurement, not a capacity bound. The torus walk is the group Z_64 x Z_64; Liu et al. (2023) show that one attention layer with an MLP can simulate such a group (their Lemmas 6 and 8), using absolute positions and width and weights growing with T, so their construction does not settle whether our 128-wide RoPE models can represent the task. Their experiments are mixed for our purpose: at T = 100 with every state labelled, the median of 20 runs stays below 90% for C_4 at 1-15 layers (and C_2, C_6 are unreliable), while C_5, C_7 and C_8 reach >= 98.5% from 3 layers. Our labels are sparse and our groups larger. On Dyck-2 at matched depth one path layer is worth about three index layers (appendix F).

Related work on depth (moved from the main text of draft 5). Shallow transformers learn automaton shortcuts unreliably (Liu et al. 2023; Hahn and Rofin 2024), while Selective RoPE (Movahedi et al.) and PaTH (Yang et al. 2025) solve parity or Flip-Flop in one layer.